

Predictive Income Modeling for Wealth Targeting

Date: July 17, 2026 | Author: Lead ML Engineer | Target: Executive Stakeholders

1. Executive Summary

This report presents our final machine learning pipeline designed to predict whether an individual's annual income exceeds \$50,000 (high earners) based on demographic and employment attributes. High-earning indicators are critical for marketing high-value financial products, optimizing advertising spend, and tailoring customer outreach. By transitioning from a baseline model to a tuned ensemble with Class Weight Balancing, we successfully boosted our ability to identify high-earning individuals (Recall) from 66.17% to 86.31% (+20.14% absolute gain) while maintaining a balanced F1-score of 0.7104. The final model has been packaged into a self-contained, fully-validated inference pipeline ready for production deployment.

2. Business Goal & Objective

- Problem:** Reaching high-income prospective clients is currently inefficient, leading to wasted marketing budget and lower conversion rates.
- Objective:** Build a classifier to identify high-income individuals (>50K) to act as the targeting engine for premium product campaigns.
- KPIs:** Maximize the identification of high earners (Recall) while maintaining a high enough Precision ($\geq 60\%$) to prevent excessive ad waste on low earners.

3. Data Profile & Feature Engineering

- Data Used:** Census Database containing 48,842 samples. Target distribution: 23.93% high-income (>50K), 76.07% low-income ($\leq 50K$).
- Features:** Selected 12 key features: 5 numeric (age, education-num, hours-per-week, capital-gain, capital-loss) and 7 categorical (workclass, marital-status, occupation, relationship, race, sex, native-country).
- Pipelines:** Numeric values are imputed with median and scaled. Categorical values are imputed with a constant missing label and one-hot encoded with ignore guards for unseen values.

4. Model Evaluation & Comparison

We evaluated three architectures on an untouched 20% hold-out test set (9,769 samples). HistGradientBoosting (HGB) was selected as the champion due to its superior F1-score and low latency:

Model	Accuracy	F1-Score	Precision	Recall	ROC AUC	PR AUC	Latency
Random Forest	86.54%	0.6779	79.31%	59.20%	0.9167	0.8060	0.0281 ms
HistGradientBoosting	87.77%	0.7214	79.29%	66.17%	0.9305	0.8350	0.0144 ms
Stacking Ensemble	87.57%	0.7156	79.12%	65.31%	0.9285	0.8325	0.0373 ms

5. Addressing Class Imbalance & Threshold Selection

By default, HGB over-optimizes accuracy on the majority class (Recall 66.17%). We evaluated class balancing strategies inside cross-validation, selecting Class Weight (Balanced) at a threshold of 0.50:

Strategy	Mean CV F1	Holdout F1	Holdout Precision	Holdout Recall	Holdout ROC AUC
Baseline (Unbalanced)	0.7214	0.7214	79.29%	66.17%	0.9305
Class Weight (Balanced)	0.7128	0.7104	60.36%	86.31%	0.9299

Random Under-Sampling	0.7056	0.7092	60.06%	86.57%	0.9277
SMOTE (Oversampling)	0.7062	0.7079	65.49%	77.03%	0.9196

6. Interpretability & Key Findings

- **Top Features:** Permutation and SHAP analysis indicate marital-status (Married-civ-spouse), capital-gain (investments), and education-num are the strongest global indicators of high income.
- **True Positive:** A married male local government employee (age 47) with an associate degree was correctly classified as >50K due to married-civ-spouse status and education.
- **False Positive:** A married male local government executive (age 53) working 50 hours/week was classified as >50K but actually earned <=50K, demonstrating segment variance.
- **False Negative:** A married male private employee (age 43) working 40 hours/week but with low education (HS-grad) was predicted as <=50K, but actually earned >50K, due to education-num pulling down the score.

7. Fairness & Bias Observations

Subgroup analysis on the hold-out test set reveals demographic and racial disparities:

Sex	Count	Accuracy	F1-Score	Precision	Recall	Selection Rate
Female	3,259	91.90%	0.6733	60.85%	75.35%	13.72%
Male	6,510	78.79%	0.7166	60.29%	88.32%	44.49%

- **Disparities:** Males are predicted as high earners at 3.2x the rate of females (44.49% vs 13.72%), and high-earning females are missed more frequently (Recall 75.35% vs 88.32%). Black and Amer-Indian subgroups also show significantly lower recalls compared to White and Asian subgroups.
- **Mitigation:** We recommend equalizing group recalls by applying group-specific threshold adjustments (Equalized Odds post-processing) before final deployment.

8. Deployment Plan & Next Steps

- **Model Artifact:** Saved as final_capstone_model.joblib. Integrated inside infer.py with schemas validation and unit tests.
- **A/B Testing:** Execute a 50/50 split test. Control: Standard marketing heuristic. Treatment: Target users predicted >50K. Measure CPA and Conversion Rate.
- **Monitoring:** Deploy drift alerts using PSI (Warning: PSI >= 0.10, Critical Action: PSI >= 0.25). Schedule model retraining semi-annually.